

How Chips Are Made: From Design to Packaging



STEP 1: DESIGN ECOSYSTEM

► EDA (Electronic Design Automation):

Use of specialised software for designing and verifying integrated circuits

► IP (Intellectual Property):

Pre-designed, reusable blocks that are integrated into chip designs to accelerate development

► Chip Design:

The process of creating detailed blueprint and layout for an integrated circuit



STEP 2: WAFER MANUFACTURING ECOSYSTEM/FRONT-END MANUFACTURING

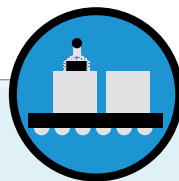
► **Ingot/Si Wafer:** Begins with the creation of silicon ingots, which are then sliced into thin uniform discs called silicon wafers, forming the base material for semiconductor devices

► Fab (Fabrication):

Core manufacturing stage. Integrated circuits are physically built onto silicon wafers using processes like deposition, etching, and photolithography

► Bump/WLT (Wafer Level Test):

Involves creating electrical connections (bumps) on the wafer and then performing initial electrical testing on individual chips while they are still on the wafer



STEP 3: CHIP PACKAGING & TESTING ECOSYSTEM/BACK-END MANUFACTURING

► **Assembly:** Individual chips (dies) are separated from wafer and prepared for packaging

► **Test:** Further rigorous testing of assembled chips to ensure they meet performance and quality specifications

► **Laser Marking:** Marking or labelling packaged chips using laser

► **QnR (Quality and Reliability):** Checks and processes to ensure quality and long-term reliability of the devices.

► **Packaging:** The process of enclosing the chip in a protective casing, providing electrical connections to the outside world, and protecting it from environmental damage.

► SMT (Surface Mount Technology):

Refers to the method of mounting packaged semiconductor components directly onto the surface of printed circuit boards. Source: Fab Economics

such as land, power supply, connectivity, and access to a skilled worker pool.”

GLIMMERS OF HOPE

India's efforts are gaining momentum. While front-end fabs are taking time to materialise, back-end manufacturing is advancing, with three chip packaging plants nearing pilot runs. For decades, Malaysia has been a hub for chip packaging and testing. India is laying the groundwork for a similar backbone.

Some leading global players are acknowledging the

efforts. Germany-based Infineon, a leader in power semiconductors, views India as rapidly emerging hub for semiconductor innovation and a strategic market. It has not committed to a fab in the region but is actively exploring manufacturing partnerships. “Recently, we announced an MoU with CDIL Semiconductors to supply silicon wafers which it will package and assemble. With this strategic partnership, we are aiming to serve Indian customers in growth sectors such as renewable energy (solar) and energy storage systems,” says Vinay Shenoy, Managing Director, Infineon.